

Atharv Vatsal

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AI/ML Engineer · Full-Stack Developer · LLMs & NLP · Computer Vision · Real-Time Systems

SUMMARY

Final-year B.Tech. CSE (AI & ML) student at VIT whose AI systems, built as a Government of Himachal Pradesh intern, are in active use by **Himachal Pradesh Police**. Builds offline, low-latency systems across on-premises LLMs, NLP, computer vision, and full-stack/desktop software, shipped as zero-setup apps for non-technical users: ~5 ms ONNX inference, ~14 FPS perception on a 4 GB GPU, and police reports processed in seconds, not 15–20 min. Also learning while building open-source Rust tools; headed a 70-member media committee (responsible for documenting and covering 227+ events).

EDUCATION

Vellore Institute of Technology (VIT)

B.Tech. in Computer Science and Engineering, Specialization in Artificial Intelligence & Machine Learning

Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, OOPS, Computer Vision, Data Structures & Algorithms, Operating Systems, DBMS

Vellore, Tamil Nadu

Expected 2027

EXPERIENCE

Department of Digital Technologies & Governance, Government of Himachal Pradesh

AI/ML Engineer Intern

Shimla, Himachal Pradesh

May 2025 – July 2025

- Delivered three end-to-end NLP, ML, and computer vision systems for public safety with HP Police, containerised with Docker.
- Contributed to **AISS**, an on-premises investigation-support system with a Deputy SP (local LLMs, Legal-BERT, OCR), now in active use.

HP Police ReportStream – NLP report processing, in active use (Python, spaCy, BERT, Mistral, PyQt, SQLite)

[GitHub](#)

- Automated extraction of 12 fields from free-text battalion daily reports sent over WhatsApp, cutting processing from **15–20 min to seconds**.
- Designed a three-tier pipeline: Regex + 10K+ typo rules (~1 s), spaCy NER + BERT (~85%), and Mistral via Ollama (~**95% accuracy**).
- Took it from a Streamlit prototype to an offline PyQt app with cross-field validation, confidence scores, batch mode, and 4 export formats.

RiskGrid – Spatio-temporal crime-risk prediction (Python, Dask, GeoPandas, XGBoost, Flask, Folium)

[GitHub](#)

- Built an ensemble (**87.4% accuracy, 89.2%** on high-risk zones) forecasting 24-hour risk in 2,847 grid cells from 7.8M+ Chicago PD records.
- Engineered 60+ spatial, temporal, and rolling features; presented a Flask heatmap and hotspot dashboard to the HP Police Cyber Cell.

Smart CCTV Initiative – State-wide, with HP Police (PyTorch, YOLO, OpenCV, GStreamer, OCR)

- Contributed YOLO vehicle detection on live RTSP CCTV streams, fine-tuned on UA-DETRAC for varied lighting, angles, and low visibility.
- Improved ALPR OCR on blurred and occluded plates; built unsafe-driving anomaly rules; explored edge deployment and thermal/IR fusion.

PROJECTS

LZ-Quant – Dual-market sentiment trading engine (PyTorch, LoRA, ONNX, FastAPI, React)

[Live Demo](#) · [GitHub](#)

- Fine-tuned DistilBERT via LoRA (296K of 66M params trainable) on ~17K texts; ONNX cut inference to **~5 ms GPU / ~55 ms CPU**.
- Streamed 10 subreddits, 6 RSS feeds, and Binance / Alpaca WebSockets (13 assets) into rolling Z-score sentiment–price divergence signals.
- Built a paper-trading engine with stop-loss, trailing stop, and a 30% drawdown circuit breaker; React dashboard for live P&L and latency.

DriveSense – Autonomous-driving perception system (PyTorch, YOLOv8, U-Net, OpenCV, CUDA)

[Demo Video](#) · [GitHub](#)

- Fused YOLOv8 detection (BDD100K, 70K images) and ResNet34 U-Net segmentation (Cityscapes): **55.6% mAP@0.5, 39.8% mean IoU**.
- Ran at **~14 FPS** (~71 ms/frame) on a 4 GB RTX 3050; evaluated 332K detections over 20K validation images with a live HUD.

keeper.raw – Offline AI photo culler, open source (Rust, Tauri, React, TypeScript, ONNX Runtime)

[GitHub](#)

- Built a cross-platform desktop app (5-crate Rust workspace + Tauri) that culls **3,000+ RAW photos in minutes** in 8 RAW formats, offline.
- Grouped bursts by time + pHash, found faces (YOLOv8-nano), scored eye sharpness, flagged blinks (MediaPipe); XMP export, 22+ tests.

A.D.A.P.T. – AI-powered UAV for flood relief, course project (Python, OpenCV, MAVLink)

- Turned satellite flood maps into drone missions: HSV segmentation, safe drop zones, TSP routing, MAVLink waypoints; tested on 3 floods.

Web Development – Next.js, TypeScript, Node.js, React, Tailwind CSS, Sanity CMS

- Built Sacred Heart School's 23-page Next.js + Sanity CMS site, editable by staff; the Matrix club site; a React portfolio.

LEADERSHIP & ACTIVITIES

VIT Vellore

Student Organizer (Head), Press & Media Committee, graVITas'26

Vellore, Tamil Nadu

2026

- Headed press, photo, video, social, podcast, and broadcasting for VIT's flagship fest (10 managers, 22 coordinators, and 38 volunteers).
- Directed coverage of **237+ events**; the committee's 300+ posts reached **4.1M+ accounts** in a month (6.7M+ in 90 days), +36% followers.
- Helped revive the graVITas YouTube channel, launched a podcast series, and coordinated print, digital, and TV media outreach.

Lead Photographer, Riviera'25 & Riviera'26 (Proshows; Lenscape Photography team)

2025 – 2026

Student Coordinator, Press & Media Committee, graVITas'25

2025

SKILLS & INTERESTS

Programming: Python, C++, Java, Rust, TypeScript, JavaScript, SQL

AI / ML: PyTorch, Scikit-learn, XGBoost, LightGBM, LLMs (Ollama, Mistral, LLaMA), LoRA, spaCy, IndicBERT, Legal-BERT

CV & Deployment: OpenCV, YOLOv8, U-Net, MediaPipe, CUDA, Tesseract OCR, ONNX Runtime, Model Quantization, Edge Inference

Backend & Apps: FastAPI, Flask, Node.js, REST APIs, WebSockets, React, Next.js, Tailwind CSS, Tauri, PyQt, SQLite, Docker, Git, Linux

Data: Pandas, NumPy, Dask, GeoPandas, Plotly, Power BI

Technical Writing: Blog on atharvvatsal.com (ML systems, building for HP Police); system blueprints for police stakeholders

Languages: English, Hindi, Punjabi, Pahadi, Haryanvi, Russian **Interests:** Event and concert photography